

Seongwon Jang

Physical AI and Robotics Engineer

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Summary

Physical AI developer experienced in structuring robot-vision problems from data collection through model design and quantitative evaluation. Built ROS2, Gazebo, YOLOv8, and multiview-learning systems for port localization, with ongoing work on Camera-LiDAR calibration, person tracking, and target reacquisition.

Technical Skills

Robotics	ROS2, Gazebo, UR5e, LeRobot, Camera-LiDAR calibration, solvePnP, multiview triangulation
Computer Vision	YOLOv8 detection/segmentation/keypoint, ONNX, VideoMAE, multiview cross attention, ReID
LLM Systems	SFT, vLLM, DeepEval Synthesizer, LLM-as-a-judge, MCP

Experience

Research Intern, Ulsan National Institute of Science and Technology Jul 2025 - Sep 2025

Algorithm research and benchmarking

- Reviewed prior algorithms, proposed and implemented a new approach, and compared results using consistent benchmarks.

Selected Projects

Future Technology Challenge 2026, CJ Logistics - Team Lead Jul 2026 - Present

Computer Vision and Robotics · [Code](#)

- Built pseudo-labeling, YOLOv8m-seg training, and ONNX conversion pipelines for monocular CCTV footage.
- Designed static false-positive removal, instance tracking, and box W/D/H estimation using a 62.3 cm rail-width reference.
- Configured an optional TensorRT FP16 execution path with engine and timing caches in ONNX Runtime.

AI for Industry Challenge - Team Lead

Apr 2026 - Present

Intrinsic and Open Robotics · ROS2 · Gazebo · UR5e · ranked 70th · [Code](#)

- Estimated port 3-DoF at approximately 5 mm MAE across five cases using YOLO keypoints and multiview triangulation.
- Reduced fine-grained XYZ MAE by 21.3% and RPY MAE by 16.5% relative to the worst case using a MultiView Bilinear Cross Attention Regressor.
- Built LeRobot/YOLO data collection with ROS timestamp checks across cameras and controller state.
- Integrated global and keypoint-local ReID with multi-camera geometry and post-occlusion target reacquisition.

RoboDine and Mobile-Robot Extension

Apr 2025 - Present

Team project and individual extension · ROS2 Jazzy · Gazebo · [RoboDine](#) · [Extension](#)

- Implemented monocular object-pose estimation using augmentation, pose-loss tuning, YOLO, and solvePnP.
- Prepared five-pose Image/CameraInfo/PointCloud2 recordings and computed Camera-LiDAR extrinsic calibration.
- Implemented person localization/tracking and closest-approach-based slowing and stopping; evaluated IDF1, ID switches, and sensor coverage in multi-person simulation.

Physical Reasoning for Structural Stability, Dacon

Mar 2026

Computer Vision - ranked 66th of 484 teams · [Code](#)

- Reduced validation log loss from 0.1216 to 0.0630 and increased accuracy from 96% to 99% using MultiView Bidirectional Cross Attention versus feature fusion.
- Reduced log loss by 86.3% over baseline using VideoMAE features and distillation regularization.

Public-Sector Q&A Chatbot, VS Community

Jan 2026 - Mar 2026

LLM Systems - Individual Project

- Improved LLM-as-a-judge results by up to 22% and response speed by up to 66% using synthetic evaluation data, SFT, asynchronous vLLM inference, and MCP integration.

Education

2019 - 2026	Bachelor’s Degree, Electronic Information and Communication Engineering , Sejong University; graduated Feb 20, 2026; GPA 4.02/4.5, major GPA 4.13/4.5
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Awards and Credentials

Jun 2025	Top Award, ROS2 and AI Autonomous Driving and Robotic Arm Bootcamp
Dec 2024	Excellence Award, IN-JEJU Challenge Hackathon
Dec 2025	Engineer Information Processing certification
Nov 2020	Class 2 Regular driver’s license (South Korea)
2021 - 2022	Republic of Korea Air Force, Information and Communications, honorably discharged as Sergeant

Languages

Korean	Native
English	OPIc IH (Mar 2026), TOEIC 925 (Jun 2025)